Anomalies

Three kinds of anomalies

- Insert
- Update
- Delete

All stem from data redundancy.

Insert anomalies caused by poor attribute grouping

student	course	room
John	COMP 140	Brockman Hall 101
Jane	COMP 140	Brockman Hall 101
John	COMP 430	Duncan Hall 1064
Mary	COMP 430	Duncan Hall 1064
Sue	COMP 430	Duncan Hall 1064
…	…	…

Need an enrolled student to reserve a room.

Assume no separate **Student** or **Course** tables.

Update anomalies caused by redundancy

student	course	room
John	COMP 140	Brockman Hall 101
Jane	COMP 140	Brockman Hall 101
John	COMP 430	Duncan Hall 1064
Mary	COMP 430	Duncan Hall 1064
Sue	COMP 430	Duncan Hall 1064
…	…	…

Updating one room results in inconsistency.

Assume no separate **Student** or **Course** tables.

Delete anomalies caused by poor attribute grouping

student	course	room
…	…	…

No enrolled students results in no information about course.

Assume no separate **Student** or **Course** tables.

What leads to such poor design?

- Spreadsheets!

- Lack of database design knowledge.
- … or just stuck using the available software.